Assignment No. 06: Kurtosis, Correlation, Rank Correlation and Regression

Blue = data from question; black = formulas. Σ in column N.

Q1. n = 250, mean 54, S.D. 3, β1 = 0, β2 = 3; items 64 and 50 were written for 62 and 52: correct constants											Σ
Given	n	250	mean	54	S.D.	3	β1	0	β2	3	
Items	wrong 1	64	wrong 2	50	right 1	62	right 2	52			
Original Σx^r	Σx	13500	Σx^2	731250	Σx^3	3.97E+07	Σx^4	2.17E+09			
Corrected Σx^r	Σx	13500	Σx^2	731202	Σx^3	3.97E+07	Σx^4	2.16E+09			
Correct	mean	54.0000	μ2	8.8080	S.D.	2.9678	β1	0.0044	β2	2.8154	

Q2. Karl Pearson's coefficient of correlation													Σ
x	25.	27.	30.	35.	33.	28.	36.						214.
y	19.	22.	27.	28.	30.	23.	28.						177.
x·y	475	594	810	980	990	644	1008						5501
x^2	625	729	900	1225	1089	784	1296						6648
y^2	361	484	729	784	900	529	784						4571
Results	n	7	r	0.8946									

Q3. Correlation coefficient: heights of fathers (x) and sons (y)													Σ
x	65.	66.	67.	67.	68.	69.	70.	72.					544.
y	67.	68.	65.	68.	72.	72.	69.	71.					552.
x·y	4355	4488	4355	4556	4896	4968	4830	5112					37560
x^2	4225	4356	4489	4489	4624	4761	4900	5184					37028
y^2	4489	4624	4225	4624	5184	5184	4761	5041					38132
Results	n	8	r	0.6030									

Q4. Proof: the correlation coefficient lies between −1 and +1 — Σ

Let $u = (x-\bar{x})/\sigma x$ and $v = (y-\bar{y})/\sigma y$, so $\Sigma u^2/n = \Sigma v^2/n = 1$ and $\Sigma uv/n = r$.

$\Sigma(u - v)^2/n \geq 0 \Rightarrow 1 + 1 - 2r \geq 0 \Rightarrow r \leq 1$; $\Sigma(u + v)^2/n \geq 0 \Rightarrow 2 + 2r \geq 0 \Rightarrow r \geq -1$; hence $-1 \leq r \leq +1$.

Check	r values: Q2 = 0.8946, Q3 = 0.6030, Q7 = 0.9500 — all within [−1, 1]: YES

Q5. Rank correlation coefficient (tied values get average ranks; tie correction applied)													Σ
X	12	15	18	20	16	15	18	22	15	21	18	15	
Y	10	18	19	12	15	19	17	19	16	14	13	17	
Rank X	12.0	9.5	5.0	3.0	7.0	9.5	5.0	1.0	9.5	2.0	5.0	9.5	
Rank Y	12.0	4.0	2.0	11.0	8.0	2.0	5.5	2.0	7.0	9.0	10.0	5.5	
d	0.0	5.5	3.0	-8.0	-1.0	7.5	-0.5	-1.0	2.5	-7.0	-5.0	4.0	
d^2	0.00	30.25	9.00	64.00	1.00	56.25	0.25	1.00	6.25	49.00	25.00	16.00	258.00
$(m^2-1)/12$ X	0.000	1.250	0.667	0.000	0.000	1.250	0.667	0.000	1.250	0.000	0.667	1.250	7.000
$(m^2-1)/12$ Y	0.000	0.000	0.667	0.000	0.000	0.667	0.250	0.667	0.000	0.000	0.000	0.250	2.500
Results	n	12	Σd^2	258.00	Σtie corr.	9.500	ρ	0.0647					

Q6. Ranks of 16 students in Mathematics and Physics: rank correlation coefficient													Σ
Maths 1-8	1	2	3	4	5	6	7	8					
Physics 1-8	1	10	3	4	5	7	2	6					
d^2	0	64	0	0	0	1	25	4					94
Maths 9-16	9	10	11	12	13	14	15	16					
Physics 9-16	8	11	15	9	14	12	16	13					
d^2	1	1	16	9	1	4	1	9					42
Results	n	16	Σd^2	136	ρ	0.8000							

Q7. Correlation coefficient and regression line y on x (and x on y); estimate y at x = 6.2													Σ
x	1.	2.	3.	4.	5.	6.	7.	8.	9.				45.
y	9.	8.	10.	12.	11.	13.	14.	16.	15.				108.
x·y	9	16	30	48	55	78	98	128	135				597
x^2	1	4	9	16	25	36	49	64	81				285
y^2	81	64	100	144	121	169	196	256	225				1356
Results	n	9	r	0.9500	b(y/x)	0.9500	a	7.2500	y = 7.2500 + 0.9500x				
Estimate	x	6.2	y est.	13.1400	b(x/y)	0.9500	a'	-6.4000	x = -6.4000 + 0.9500y				

Q8. Var x = 9; regression lines 8x − 10y + 66 = 0 and 40x − 18y = 214 (written as ax + by = c): means, σy, r												Σ
Lines	a1	8	b1	-10	c1	-66	a2	40	b2	-18	c2	214	
Given	Var x	9	*Line 1 taken as y on x, line 2 as x on y ($r^2 = b_{yx} \cdot b_{xy} \leq 1$ confirms the choice)*										
Results	mean x	13.0000	mean y	17.0000	b(y/x)	0.8000	b(x/y)	0.4500	r	0.6000	S.D. y	4.0000	

Q9. Acute angle θ between the two regression lines: $\tan\theta = ((1-r^2)/r)\cdot\sigma x\sigma y/(\sigma x^2+\sigma y^2)$ — Σ

Lines: $y-\bar{y} = r(\sigma y/\sigma x)(x-\bar{x})$ [slope m1 = rσy/σx] ; $x-\bar{x} = r(\sigma x/\sigma y)(y-\bar{y})$ [slope m2 = σy/(rσx)]. $\tan\theta = |(m2-m1)/(1+m1m2)|$

$\Rightarrow \tan\theta = |\sigma y/(r\sigma x) - r\sigma y/\sigma x| / (1 + \sigma y^2/\sigma x^2) = ((1-r^2)/r)\cdot\sigma x\sigma y/(\sigma x^2+\sigma y^2)$.

Interpretation: $r = 0 \Rightarrow \theta = 90°$ (lines perpendicular, parallel to the axes); $r = \pm 1 \Rightarrow \tan\theta = 0 \Rightarrow \theta = 0°$ (the two lines coincide).

Illustration	σx	3	σy	4		
r	0.0	0.2	0.4	0.6	0.8	1.0
θ (degrees)	90.00	66.54	45.23	27.11	12.19	0.00

Q10. Most likely Bombay price (y) for Calcutta price (x) = ₹70												Σ
Given	mean x	65	mean y	67	S.D. x	2.5	S.D. y	3.5	r	0.8	x	70
Results	b(y/x)	1.1200	y at x	72.60	y − 67 = 1.12(x − 65)							